

DM545/DM871 – Linear and integer programming

Exercise Sheet 3, Autumn 2026

Solution:

Included.

Exercises with the symbol $+$ are to be done at home before the class. Exercises with the symbol $*$ will be tackled in class and should be at least read at home. The remaining exercises are left for self training after the exercise class.

Exercise 1⁺

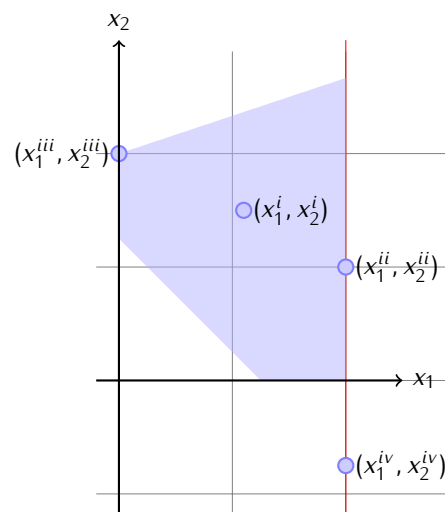
What argument is used to prove that the simplex algorithm always terminates in a finite number of iterations if it does not encounter a situation in which one of the basic variables is zero? What may happen instead if the latter situation arises and which remedies are introduced?

Exercise 2^{*}

Exercise 3 from Exam 2013.

Consider the following LP problem:

$$\begin{aligned}
 (P) \quad & \max \quad z = x_1 + 5x_2 \\
 & \text{s.t.} \quad -x_1 + 3x_2 \leq 6 \\
 & \quad \quad 4x_1 + 4x_2 \geq 5 \\
 & \quad \quad 0 \leq x_1 \leq 2 \\
 & \quad \quad x_2 \geq 0
 \end{aligned}$$



(Note: the following subtasks can be carried out independently; use fractional mode for numerical calculations)

- a. The polyhedron representing the feasibility region is depicted in the figure. Indicate for each of the four points represented whether they are feasible and/or basic solutions. Justify your answer.

Solution:

- Point 1 is a feasible solution but not basic (no constraint is active in that point).
- Point 2 is a feasible solution but not basic (only one constraint is active while two are needed)
- Point 3 is a basic feasible solution
- Point 4 is a basic solution (combination of two active constraints) but non feasible.

- b. Write the initial tableau or dictionary for the simplex method. Write the corresponding basic solution and its value. State whether the solution is feasible or not and whether it is optimal or not.

Solution:

$$\begin{aligned}
 z = \max \quad & x_1 + 5x_2 \\
 \text{s.t.} \quad & -x_1 + 3x_2 \leq 6 \\
 & -4x_1 - 4x_2 \leq -5 \\
 & x_1 \leq 2 \\
 & x_1 \geq 0 \\
 & x_2 \geq 0
 \end{aligned}$$

	x1	x2	x3	x4	x5	-z	b
I	-1	3	1	0	0	0	6
II	-4	-4	0	1	0	0	-5
III	1	0	0	0	1	0	2
IV	1	5	0	0	0	1	0

The basic solution is $x_1 = 0$, $x_2 = 0$, $x_3 = 6$, $x_4 = -5$ and $x_5 = 2$. Its value is 0. The solution is not feasible.

- c. Consider the following tableau:

	x1	x2	x3	x4	x5	-z	b
I	0	4	1	-1/4	0	0	29/4
II	1	1	0	-1/4	0	0	5/4
III	0	-1	0	1/4	1	0	3/4
IV	0	4	0	1/4	0	1	-5/4

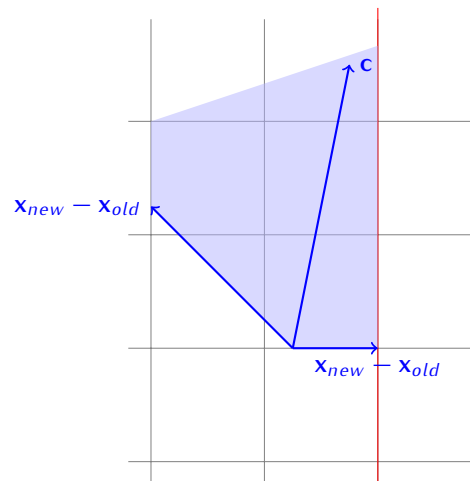
and the following three pivoting rules:

- largest coefficient
- largest increase
- steepest edge.

Which entering and leaving variables would each of them indicate? In this specific case, which rule would be convenient to follow? Report the details of the computations for the first two rules and carry out graphically the application of the third rule using the plot in the figure above (tikz code to reproduce the figure available in the online version.)

Solution:

- The two candidate entering variables are x_2 and x_4 . The reduced cost of x_2 is larger hence that is the entering variable. The leaving variable is consequently given by the ratio test and is x_1 since $5/4 < 29/16$.
- The two candidate entering variables are x_2 and x_4 . The increase possible with x_2 is $\min\{29/4 \cdot 1/4, 5/4 \cdot 1\} \cdot 4 = 5$ while the increase with x_4 is $\min\{3/4 \cdot 4\} \cdot 1/4 = 3/4$. Hence x_2 is the entering variable and the leaving variable is x_1 .
- In the figure we plot the vector $\mathbf{c}$ which is the perpendicular to the objective function and the two vectors corresponding to the movement we would take by the iteration of the simplex. The angle between $\mathbf{c}$ and $\mathbf{x}_{new} - \mathbf{x}_{old}$ is smaller for the decision x_2 entering x_1 leaving.



None of the three rules is convenient, the best would be to let x_4 enter and x_5 leave, we would reach the optimal solution in less iterations.

Exercise 3

The two following LP problems lead to two particular cases when solved by the simplex algorithm. Identify these cases and characterize them, that is, give indication of which conditions generate them in general.

$$\begin{array}{ll} \text{maximize} & 2x_1 + x_2 \\ \text{subject to} & x_2 \leq 5 \\ & -x_1 + x_2 \leq 1 \\ & x_1, x_2 \geq 0 \end{array}$$

$$\begin{array}{ll} \text{maximize} & x_1 + x_2 \\ \text{subject to} & 5x_1 + 10x_2 \leq 60 \\ & 4x_1 + 4x_2 \leq 40 \\ & x_1, x_2 \geq 0 \end{array}$$

Solution:

In the initial tableau of the first problem there is a column, the first one, with positive reduced cost and no positive a_{ij} term. This means that the corresponding variable x_1 can be brought into the basis but the increase of its value is unlimited. This indicates that we have an unbounded problem.

The second LP problem is developed in the slides for the lecture on exception handling. After some iterations we reach a tableau in which a non basic variable has reduced cost zero. This indicates that it can be brought in the basis without a change in the objective function. Since the solution changes when we bring the variable in basis then the problem has more than one solution and it has therefore infinite solutions. They can be expressed as the convex combination of all optimal basic solutions.

Exercise 4*

Consider the following problem:

$$\begin{array}{ll} \max & z = 4x_2 \\ \text{s.t.} & 2x_2 \geq 0 \\ & -3x_1 + 4x_2 \geq 1 \\ & x_1, x_2 \geq 0 \end{array}$$

- Write the LP in equational standard form and say why it does not provide immediately an initial feasible basis for the simplex method.

Solution:

In the equational standard form we have a negative b term. The implication of this is that the initial solution of the simplex is infeasible because $x_B \not\geq 0$. If we try to make the term positive we end up not having an identity matrix in the tableau.

Project	Year 1	Year 2	Year 3	Year 4	Year 5
1	-1.00	0.50	0.30	1.80	1.20
2	-1.00	0.60	0.20	1.50	1.30
3	0.00	-1.00	0.80	1.90	0.80
4	-1.00	0.40	0.60	1.80	0.95

- b. To overcome the situation of infeasible basis construct the auxiliary problem for a phase I-phase II solution approach. Determine which variables are initially in basis and which are not in basis in the auxiliary problem.
- c. Answer the following questions
- Is the initial basis in the auxiliary problem feasible in the original problem?
 - Is it optimal in the auxiliary problem?
 - Is it degenerate?
 - Can we say at this stage if phase I will terminate?
 - If it will terminate, can we say at this stage that it will terminate with a basis that corresponds to a feasible solution in the original problem?
 - Solve the problem by carrying out Phase I and Phase II of the simplex algorithm.

Exercise 5⁺ LP modeling — Investment plan

An investor has 10,000 Dkk to invest in four projects. The following table gives the cash flow for the four investments.

The information in the table can be interpreted as follows: For project 1, 1.00 Dkk invested at the start of year 1 will yield 0.50 Dkk at the start of year 2, 0.30 Dkk at the start of year 3, 1.80 Dkk at the start of year 4, and 1.20 Dkk at the start of year 5. The remaining entries can be interpreted similarly. The entry 0.00 indicates that no transaction is taking place. The investor has the additional option of investing in a bank account that earns 6.5% annually. All funds accumulated at the end of 1 year can be reinvested in the following year. Formulate the problem as a linear program to determine the optimal allocation of funds to investment opportunities.

[Taken from Operations Research: An Introduction, Taha]

Solution:

Let

x_i = Krone invested in project i , $i = 1, 2, 3, 4$

y_j = Krone invested in bank in year j , $j = 1, 2, 3, 4$

$$\max z = y_5$$

$$x_1 + x_2 + x_4 + y_1 \leq 10000$$

$$.5x_1 + .6x_2 - x_3 + .4x_4 + 1.065y_1 - y_2 = 0$$

$$.3x_1 + .2x_2 + .8x_3 + .6x_4 + 1.065y_2 - y_3 = 0$$

$$1.8x_1 + 1.5x_2 + 1.9x_3 + 1.8x_4 + 1.065y_3 - y_4 = 0$$

$$1.2x_1 + 1.3x_2 + .8x_3 + .95x_4 + 1.065y_4 - y_5 = 0$$

$$x_1, x_2, x_3, x_4, y_1, y_2, y_3, y_4, y_5 \geq 0$$

Optimal solution:

$$x_1 = 0, x_2 = 10,000, x_3 = 6000, x_4 = 0$$

$$y_1 = 0, y_2 = 0, y_3 = 6800, y_4 = 33,642$$

$$z = 53,628.73 \text{ at the start of year 5}$$

Exercise 6* LP modeling — Budget Allocation

A company has six different opportunities to invest money. Each opportunity requires a certain investment over a period of 6 years or less. See Figure 1.

The company wants to invest in those opportunities that maximize the combined *Net Present Value* (NPV). It also has an investment budget that needs to be met for each year. (The Net Present Value is calculated with an interest rate of 5%).

How should the company invest?

We assume that it is possible to invest partially in an opportunity. For instance, if the company decides to invest 50% of the required amount in an opportunity, the return will also be 50%.

Net present value:

A debtor wants to delay the payment back of a loan for t years. Let P be the value of the original payment presently due. Let r be the market rate of return on a similar investment asset. The future value of P is

$$F = P(1 + r)^t$$

Viceversa, consider the task of finding the present value P of \$100 that will be received in five years, or equivalently, which amount of money today will grow to \$100 in five years when subject to a constant discount rate. Assuming a 5% per year interest rate, it follows that

$$P = \frac{F}{(1 + r)^t} = \frac{\$100}{(1 + 0.05)^5} = \$78.35.$$

Solution:

Net Present Value calculation:

for each opportunity we calculate the NPV at time zero (the time of decision) as:

$$P_0 = \sum_{t=1}^5 \frac{F_t}{(1 + 0.05)^t}$$

Let B_t be the budget available for investments during the years $t = 1..5$. Let a_{tj} be the cash flow for opportunity j and c_j its NPV. We want to choose a set of opportunities such that the budget is never exceeded and the expected return is maximized. We consider divisible opportunities.

Variables $x_j \in \mathbb{R}^+$ the quantity of opportunity j selected, $j = 1..6$

Objective

$$\max \sum_{j=1}^6 c_j x_j$$

Constraints

$$\sum_{j=1}^6 a_{tj} x_j + B_t \geq 0 \quad \forall t = 1..5$$

Expected Investment Cash Flows and Net Present Value							
	Opp. 1	Opp. 2	Opp. 3	Opp. 4	Opp. 5	Opp. 6	Budget
Year 1	-\$5.00	-\$9.00	-\$12.00	-\$7.00	-\$20.00	-\$18.00	\$45.00
Year 2	-\$6.00	-\$6.00	-\$10.00	-\$5.00	\$6.00	-\$15.00	\$30.00
Year 3	-\$16.00	\$6.10	-\$5.00	-\$20.00	\$6.00	-\$10.00	\$20.00
Year 4	\$12.00	\$4.00	-\$5.00	-\$10.00	\$6.00	-\$10.00	\$0.00
Year 5	\$14.00	\$5.00	\$25.00	-\$15.00	\$6.00	\$35.00	\$0.00
Year 6	\$15.00	\$5.00	\$15.00	\$75.00	\$6.00	\$35.00	\$0.00
NPV	\$8.01	\$2.20	\$1.85	\$7.51	\$5.69	\$5.93	

Figure 1: